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Summary

Computer Science (AI) student with hands-on experience building end-to-end machine learning and LLM-powered systems: from classical ML pipelines to a production RAG platform and multi-agent AI simulations. Comfortable across the full ML lifecycle – data preprocessing, model training/evaluation, and deployment behind a real backend.

Featured AI/ML Projects

Suisik – AI-Powered Personalized Learning Platform

RAG / LLM Application — Next.js, Django, Celery, LLMs (Mistral/Llama)

- Built a Retrieval-Augmented Generation pipeline that parses uploaded PDF/DOCX study material into hierarchical chunks and topic graphs
- Implemented LLM-based topic extraction and an adaptive quiz-generation engine producing MCQ, multi-select, and true/false questions tailored to student proficiency
- Designed async processing with Celery and a Django REST/Channels backend serving a Next.js frontend

AI-TASK – Risk-Aware Multi-Agent Pathfinding

Academic AI Coursework — Python, A Search, Probabilistic Reasoning*

- Designed a multi-agent system where four robots navigate a partially observable 15x15 grid with hidden mines, using A* search combined with local probabilistic risk estimation
- Built a short-horizon collision-avoidance layer and an experiment harness producing 50+ statistical figures (win rate, survival rate, efficiency) across configurations and seeds

Customer Churn Predictor

Supervised Learning — Python, XGBoost, TensorFlow, Scikit-learn

- Built and compared XGBoost and neural network classifiers to predict customer churn from structured data
- Engineered preprocessing pipelines and evaluated feature importance to improve model reliability

Models From Scratch

Python, NumPy, Matplotlib

- Implemented Linear Regression, Logistic Regression, and a Decision Tree Classifier from first principles using only NumPy – no scikit-learn

Used Car Price Prediction

Python, Pandas, Scikit-learn, Seaborn

- Built a regression model estimating used car prices from engine size, mileage, and brand, with a market-segmentation analysis (mainstream vs. luxury) and a written technical report

Image Captioning System

Jupyter/Python, InceptionV3, LSTM with Attention

- Combined InceptionV3 feature extraction with an attention-based LSTM decoder to generate natural-language captions for images from the MS COCO dataset

Technical Skills

- **Languages:** Python, SQL
- **ML/DL:** TensorFlow, Keras, Scikit-learn, XGBoost, NumPy, Pandas
- **LLM/AI Systems:** RAG pipelines, LLM integration (Mistral, Llama via Ollama), vector embeddings, prompt-driven quiz/content generation
- **Backend:** Django, Django REST Framework, Celery, Django Channels
- **Tools:** Jupyter, Git, Google Colab, Kaggle

Certifications

- Diploma in Data Science and Machine Learning – Vrit Technologies
- LangChain + Gemini for LLM Applications
- Natural Language Processing with HuggingFace Transformers

Education

Softwarica College of IT and E-Commerce, Kathmandu
Bachelor's in Computer Science (Artificial Intelligence focus)